Danielle Pauleen G. Labasa BSIT 2-2N object-Oriented Programming 2-27-2026

Part A. Variable Identification Table

Variable	Data Type	Variable type	Stored In
school	String	static	memory
name	String	instance	heap
grade	int	instance	heap
gwa	double	local	stack
s1	Student	reference	stack
s2	Student	reference	stack

Part B. Explanation

Statement	What Happens?	Memory Impact
s1 = newStudent ();	New object is created into the heap. The variable points to the object created.	one Student object is created in the heap. s1 stores the reference to the object in the stack.
s2 = new Student ();	Another object is created. The variable points to a different object now.	two Student objects now exist in the heap, then s2 points to the 2nd student object in stack.
s2 = s1	s2 is now the same object as s1. They have the same value. The previous value is trashed.	s1 and s2 now points to the same object in the stack. The second object can be garbaged in the heap.
s1 = null	s1 points to null.	s1 points to null in the stack, but first obj still exists in the heap.
school = "PUP Sta. Mesa"		Updates all student obj as it is from method area

Part C. Output

Mark 90 1.75 PUP-Sta. Mesa (-assignment happened before s1 null -s2 points to the first student object)